Modeling Radiotrophic Fitness

Spatiotemporal Dynamics of Microbial Communities Under Ionizing Radiation Stress

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Abstract

This work presents a mathematical framework to describe the radiotrophic and adaptive dynamics of microbial communities exposed to gamma radiation. The developed model integrates Langevin dynamics with reaction-diffusion equations to simulate the spatial motility, growth, and nonlinear interactions of radiotrophic fungi, extremophiles, and radiation-sensitive averse species. In this framework, radiation gradients and chemical factors, such as melanin synthesis, are introduced as modifiable parameters to analyze their influence on biofilm growth and adaptive responses. The model leverages stochastic differential equations to capture complex dependencies between environmental conditions and microbial interactions. Numerical simulations of seven interacting species—including *Cryptococcus neoformans*, *Deinococcus radiodurans*, *Cladosporium sphaerospermum*, *Bacillus subtilis*, *Aspergillus niger*, *Shewanella oneidensis*, and *Ochrobactrum intermedium* AM7—demonstrate that radiation gradients drive spatial niche partitioning, with melanized radiotrophic fungi concentrating in high-radiation zones while radiosensitive metal-reducing bacteria occupy attenuated peripheries. This work contributes a predictive modeling tool capable of elucidating the nonlinear adaptive behaviors of radiotrophic communities in extreme environments, with applications to nuclear bioremediation.

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1. Introduction

Microbial communities possess an extraordinary ability to adapt and survive under extreme environmental stressors, such as elevated radiation levels. Within such communities, radiotrophic fungi and other extremophilic microorganisms have evolved specialized mechanisms that allow them to thrive in high-radiation environments. This study aims to develop a comprehensive mathematical framework that captures the complex interplay between these organisms' nonlinear interactions, radiation sensitivities, and chemotactic behaviors within biofilm communities. By leveraging a combined approach that integrates Langevin dynamics, reaction-diffusion modeling, and a Hamiltonian-based formulation, we simulate and explore the spatiotemporal evolution of these communities under radiation stress and other external influences.

The practical motivation for this work lies in nuclear bioremediation. Contaminated sites such as the Chernobyl Exclusion Zone and the Hanford Nuclear Reservation host complex microbial communities in which radiotrophic species coexist with metal-reducing bacteria capable of immobilizing radionuclides. Mathematical models that predict community dynamics under spatially heterogeneous radiation fields are essential for designing and optimizing bioremediation strategies. The framework developed here provides the theoretical foundation for such predictions.

2. Related Work

2.1 Radiotrophic Organisms and Melanin-Mediated Radiotrophy

The discovery that melanized fungi thrive in high-radiation environments has fundamentally reshaped our understanding of biological energy transduction. Zhdanova et al. [1] first documented the prevalence of darkly pigmented fungi within the damaged reactor at the Chernobyl Nuclear Power Plant, noting that species such as *Cladosporium sphaerospermum* dominated the mycobiota of the most contaminated structures. Subsequent work by Dadachova et al. [2] demonstrated that ionizing radiation alters the electronic properties of melanin, enhancing electron-transfer activity in melanized cells of *Cryptococcus neoformans* and *Wangiella dermatitidis*. Melanized cells exposed to radiation levels approximately 500 times above background grew significantly faster, accumulated more biomass, and incorporated three-fold more ^{14}C-acetate than non-irradiated melanized cells or irradiated albino mutants. Dadachova and Casadevall [3] subsequently proposed the term "radiosynthesis" to describe this phenomenon, drawing an analogy to photosynthesis in which melanin serves a role loosely comparable to chlorophyll. More recently, Shunk et al. [4] cultivated *C. sphaerospermum* aboard the International Space Station for 26 days, demonstrating a growth advantage of approximately 21% under space radiation conditions and measurable attenuation of ionizing radiation beneath the fungal biomass.

The radiation resistance of *Deinococcus radiodurans* operates through fundamentally different mechanisms. Rather than harnessing radiation for metabolic benefit, *D. radiodurans* survives doses exceeding 12 kGy through a combination of efficient DNA double-strand break repair via extended synthesis-dependent strand annealing (ESDSA), multiple genome copies enabling homologous recombination, and a potent manganese-based antioxidant system that protects proteins from oxidative damage [5, 6]. Daly et al. [7] showed that small-molecule Mn^{2+}-metabolite complexes specifically protect the proteome against radiation-induced carbonylation, establishing that protein protection, rather than DNA repair capacity alone, governs extreme radioresistance. *Aspergillus niger*, while not radiotrophic in the strict sense, produces melanin that confers substantial radioprotection, with melanized fungi generally exhibiting LD_{10} values approaching or exceeding 1 kGy [3].

2.2 Mathematical Models of Biofilm Communities

The mathematical modeling of biofilm communities has a rich history beginning with the foundational one-dimensional multispecies model of Wanner and Gujer [8], which coupled reaction-diffusion equations for substrate transport with biomass conservation laws to predict

biofilm thickness and spatial species distributions. Xavier et al. [9] extended this framework to multiple spatial dimensions, providing a deterministic continuum approach to heterogeneous biofilm development. Individual-based modeling approaches, exemplified by the iDynoMiCS platform of Lardon et al. [10], complement continuum models by resolving cell-level stochastic interactions. Cross-diffusion biofilm models have been analyzed by Sonner et al. [11], who established mathematical properties of volume-filling multispecies systems that exhibit porous-medium-type degeneracy. Our PSDE framework extends these precedents by incorporating radiation-dependent fitness terms and phase-locking dynamics that couple species interactions to external radiation fields, moving beyond the substrate-limited growth paradigms that dominate existing biofilm models.

2.3 Hamiltonian and Stochastic Approaches in Theoretical Ecology

The application of Hamiltonian mechanics to ecological systems remains comparatively underexplored, despite the natural conservation laws that govern energy flow in ecosystems. Symplectic integration methods, which preserve the geometric structure of Hamiltonian flows, have been advocated for ecological modeling by Diele et al. [12] to ensure that numerical solutions maintain essential qualitative dynamics. The Kuramoto model of coupled phase oscillators [13] provides a well-established framework for synchronization phenomena in biological systems, from neural networks to circadian rhythms. Acebŕón et al. [14] reviewed the model as a paradigm for synchronization, noting its generalization to systems with time delays, frequency-weighted coupling, and heterogeneous interaction topologies. Our phase-locking kernel $\Gamma_s(t, \mathbf{x})$ draws directly on this tradition, extending it to multispecies microbial communities where species-level oscillatory dynamics in growth and resource acquisition couple through shared radiation and nutrient fields. The use of Langevin stochastic dynamics to describe microbial population fluctuations follows naturally from the Fokker-Planck formalism and has been applied to competing microbial populations in chemostat systems [15], where stochastic differential equations revealed long-term competitive outcomes differing from deterministic predictions.

2.4 Bioremediation Context

The practical motivation for modeling radiotrophic communities lies in nuclear bioremediation. *Shewanella oneidensis* MR-1 is a model dissimilatory metal-reducing bacterium capable of reducing soluble U(VI) to insoluble U(IV) via extracellular electron transfer through c-type cytochromes [16, 17]. *Ochrobactrum intermedium* AM7, isolated from soil near the Kakrapar Atomic Power Station in India, tolerates high concentrations of thorium and produces exopolysaccharides (EPS) that complex with Th(IV), offering a pathway toward bioremediation of actinide-contaminated environments [18]. The combination of radiotrophic fungi that actively benefit from radiation with metal-reducing bacteria that immobilize radionuclides suggests the possibility of engineered biofilm consortia for in situ remediation. The fitness landscape framework introduced by Wright [19] and formalized through the NK model of Kauffman [20] provides a conceptual foundation for understanding how epistatic interactions among species traits shape adaptation under radiation stress, a perspective that our Hamiltonian kNN decision tree operationalizes quantitatively.

3. Mathematical Framework

3.1 Symbols and Notation

Table 1: Summary of principal mathematical symbols.

Symbol	Description
$F_s(t, \mathbf{x})$	Fitness function of species s, evolving over time t and space $\mathbf{x}$
$H(p, q)$	Hamiltonian function defining total system energy
p, q	Phase-space variables (generalised momentum and position)
$\eta_s(t, \mathbf{x}) = \sigma_s \xi(t, \mathbf{x})$	Stochastic noise: σ_s intensity; ξ spatiotemporal white noise
D_s	Species-specific passive diffusion coefficient
μ_s	Species-specific directed-motility (chemotaxis) coefficient
$P_{sj}(t)$	Permutation matrix describing species-s to species-j transitions
$I_\gamma(t, \mathbf{x})$	Ionizing (gamma) radiation field
$N(t, \mathbf{x})$	Non-ionizing (UV) radiation field
k_B	Boltzmann constant
$S[q(t)]$	Action functional for species trajectories
∇	Spatial gradient operator
$\Gamma_s(t, \mathbf{x})$	Phase-locked kernel for species s
$\beta_{s,\text{ion}}$	Ionizing radiation sensitivity coefficient
$\alpha_{s,\text{nir}}$	Non-ionizing radiation coupling coefficient
θ_s	Adaptive phase-locking coefficient
γ_s	Positive gamma-radiation sensitivity (benefit) coefficient
Δ_s	Phase-locking synchronization deviation

3.2 Partial Stochastic Differential Equation (PSDE) for Species Fitness

The fitness of each species s is governed by the following PSDE:

$$\begin{aligned}\partial_t F_s = \nabla \cdot (D_s \nabla F_s) - \nabla \cdot \left(\mu_s \sum_{j=1}^{n} P_{sj}(t)\, F_j \right) + R_s + \sigma_s \xi(t, \mathbf{x}) \\ - \beta_{s,\text{ion}}\, I_\gamma F_s + \gamma_s \Delta_s - \alpha_{s,\text{nir}}\, N F_s + \theta_s H_s + C_s \end{aligned} \tag{1}$$

where $F_s \equiv F_s(t, \mathbf{x})$, $I_\gamma \equiv I_\gamma(t, \mathbf{x})$, and $N \equiv N(t, \mathbf{x})$. The nine terms encode:

- **Diffusion**: passive spatial spreading at rate D_s.
- **Directed motility**: chemotaxis and radiation-gradient-directed movement controlled by μ_s and the transition matrix P_{sj}.
- **Deterministic reaction**: $R_s(t, \mathbf{x})$, nutrient-dependent growth (see Section 3.7).
- **Stochastic noise**: $\sigma_s \xi(t, \mathbf{x})$, spatiotemporal white noise scaled by species noise intensity σ_s.
- **Ionising damage**: $-\beta_{s,\text{ion}}\, I_\gamma(t, \mathbf{x})\, F_s$, radiation-induced fitness reduction.
- **Phase-locking adjustment**: $\gamma_s \Delta_s$, adaptive synchronisation gain from melanin-mediated energy transduction.

- **Non-ionising effect**: $-\alpha_{s,\text{nir}}\, N(t,\mathbf{x})\, F_s$, UV/heat-driven fitness coupling.
- **Hamiltonian interaction**: $\theta_s H_s(t,\mathbf{x})$, energy-conserving inter-species force from the Hamiltonian (Section 3.3).
- **External forcing**: $C_s(t,\mathbf{x})$, environmental control inputs.

3.3 Hamiltonian Framework

The total Hamiltonian for the multi-species biofilm is:

$$H = \sum_{i=1}^{N} \left[\frac{1}{2}\, \rho_i\, v_i^2 + U_i(\mathbf{x}_i) \right] + \sum_{i \neq j} V_{ij}(r_{ij}) + \sum_{k=1}^{K} W_k(t,\mathbf{x}) \tag{2}$$

where ρ_i is the effective biomass density of species i [cells μm^{-3} $\times$ effective mass analog], $U_i(\mathbf{x}_i)$ is the potential energy in the nutrient concentration field, $V_{ij}(r_{ij})$ is the pairwise interaction energy, and $W_k(t,\mathbf{x})$ captures external energy contributions from radiation exposure and stochastic effects. For mutualistic interactions:

$$V_{ij}^{\text{mutual}}(r_{ij}) = -\gamma\, \exp\left(-\frac{r_{ij}^2}{\sigma^2} \right) \tag{3}$$

3.4 Symplectic Integration

Symplectic integrators preserve the geometric structure of the Hamiltonian flow, ensuring that energy and momentum do not drift over extended simulation windows. Phase-space trajectories evolve via the phase-locked canonical equations:

$$\frac{dq}{dt} = \frac{\partial H}{\partial p} - \Gamma_s(t,\mathbf{x})\, F_s(t,\mathbf{x}), \qquad \frac{dp}{dt} = -\frac{\partial H}{\partial q} + \Gamma_s(t,\mathbf{x})\, F_s(t,\mathbf{x}) \tag{4}$$

The leapfrog/Verlet scheme is employed numerically: (1) evaluate H_s from the current state; (2) adjust the phase-locking deviation Δ_s; (3) advance via a symplectic step [33].

3.5 Radiation Field Models

Ionizing radiation (gamma decay).

$$I_\gamma(t,\mathbf{x}) = I_0\, \exp(-\kappa\, \|\mathbf{x}\|) \tag{5}$$

where I_0 is the initial intensity at the source and κ is the attenuation coefficient scaled by local biofilm density.

Non-ionizing radiation (UV oscillatory field).

$$N(t,\mathbf{x}) = N_{\text{UV}} \cos(\omega t) \tag{6}$$

3.6 Melanin Reaction-Diffusion

The spatial production and diffusion of melanin by radiotrophic fungi is modelled as:

$$\frac{\partial M}{\partial t} = D_M \, \nabla^2 M + \alpha_M \cdot n_{\mathrm{RF}}(\mathbf{x}, t) \cdot I_\gamma(t, \mathbf{x}) \tag{7}$$

where D_M is the melanin diffusion coefficient [$\mu\mathrm{m}^2\,\mathrm{s}^{-1}$], α_M is the radiation-driven production rate [μg cell^{-1} Gy^{-1}], and n_{RF} is the local radiotrophic cell density [cells $\mu\mathrm{m}^{-3}$]. Each term carries units of [$\mu\mathrm{g}\,\mu\mathrm{m}^{-3}\,\mathrm{s}^{-1}$].

3.7 Nutrient Uptake with Adaptive Feedback

$$R_s(t, \mathbf{x}) = \alpha_s \, C_s(t, \mathbf{x}) \left(1 - \frac{F_s(t, \mathbf{x})}{K_s}\right) + A_s(t) \, C_s(t, \mathbf{x}) \tag{8}$$

$$A_s(t) = \gamma_s \, \phi_s(t) - \beta_{s,\mathrm{ion}} \, I_\gamma(t) \tag{9}$$

where $\phi_s(t) = \cos(\omega_s t - \theta_s)$ is the phase-alignment function and $A_s(t)$ is the adaptive feedback driven by phase-locking and local radiation stress.

3.8 Hamiltonian kNN Decision Tree

To model species transition probabilities modulated by neighbourhood phase states:

$$H_{\mathrm{kNN}} = \frac{1}{\sigma_n} \sum_{j=1}^{n} P_{sj}(t) \, F_j(t, \mathbf{x}) \, \Gamma_s(t, \mathbf{x}) \tag{10}$$

3.9 Radiation Damage and Resilience

$$D_{\mathrm{rad},s}(t, \mathbf{x}) = -\beta_{s,\mathrm{rad}} \, I_\gamma(t, \mathbf{x}) \, F_s(t, \mathbf{x}) \tag{11}$$

Species with higher melanin production or stronger DNA repair exhibit lower $\beta_{s,\mathrm{rad}}$ values, reflecting the Mn^{2+}-dependent proteome protection described by Daly et al. [7].

3.10 Biofilm Mechanical Properties

The EPS matrix is modelled as a Finite Extensible Nonlinear Elastic (FENE) entropic spring:

$$U(r) = -\frac{1}{2} \, k \, R^2 \, \ln\left(1 - \frac{r^2}{R^2}\right) \tag{12}$$

where k is the spring constant, R is the maximum extensibility, and r is the current end-to-end distance. Viscoelastic creep follows the Kelvin-Voigt model:

$$\sigma(t) = E \, \varepsilon(t) + \eta \, \frac{d\varepsilon}{dt} \tag{13}$$

with stress relaxation time $\tau = \eta / E$ governing biofilm restructuring dynamics.

3.11 Radiodialysis Membrane Transport

To model contaminant ingress through the bioreactor membrane under radiation-driven permeability change, we introduce a minimal three-equation system coupling mobile contaminant transport, biosorption/bioreduction, and membrane structural damage.

Mobile contaminant — cylindrical reaction-diffusion. The concentration $c(r,t)$ of a dissolved contaminant (e.g. aqueous U(VI), Cs^+) in the cylindrical biofilm domain satisfies:

$$\frac{\partial c}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\, D_{\text{eff}}\frac{\partial c}{\partial r}\right) - \left(k_{\text{ads}}X + k_{\text{red}}X_{\text{red}}\right)c + k_{\text{des}}\, s \tag{14}$$

where D_{eff} is the effective diffusivity through the EPS matrix, k_{ads} and k_{red} are the biosorption and bioreduction rate constants, X and X_{red} are the total and metal-reducing biomass densities, and $s(r,t)$ is the immobilised contaminant concentration (sorbed or reduced phase). Zero-flux symmetry is imposed at $r = 0$.

Immobile phase — biosorption and bioreduction.

$$\frac{\partial s}{\partial t} = \left(k_{\text{ads}}X + k_{\text{red}}X_{\text{red}}\right)c - \left(k_{\text{des}} + k_{\text{loss}}\right)s \tag{15}$$

where k_{loss} accounts for precipitation, co-precipitation, or export of the immobilised fraction [42].

Membrane damage and radiation-driven permeability. Radiation progressively degrades membrane structural integrity $m(t) \in [0,1]$:

$$\frac{dm}{dt} = -k_{\text{dam}}\,\dot{D}(R)\,m, \qquad m(0) = 1 \tag{16}$$

where $\dot{D}(R)$ is the absorbed dose rate at the membrane wall ($r = R$). The effective permeability P_{eff} evolves with cumulative dose $D_{\text{cum}}(t) = \dot{D}(R)\cdot t$:

$$P_{\text{eff}}(t) = P_0\,\exp\!\big(\alpha_P\, D_{\text{cum}}(t)\big) \tag{17}$$

consistent with radiation-driven permeability enhancement observed in Nafion membranes under ionising flux [41].

Robin boundary condition at $r = R$. The membrane couples interior concentration gradient to the external reservoir c_{ext}:

$$-D_{\text{eff}}\left.\frac{\partial c}{\partial r}\right|_{r=R} = P_{\text{eff}}(t)\left(c(R,t) - c_{\text{ext}}\right) \tag{18}$$

This is analogous to Donnan dialysis through a charged membrane [43, 40], with the critical distinction that P_{eff} is not a fixed material property but an evolving field driven by cumulative

radiation exposure — a self-regulating mechanism in which the radiation that sustains the biofilm simultaneously widens the pathway for contaminant uptake.

4. Parameter Estimation

Biologically justified parameter ranges for the six primary modelled species, derived from published experimental data, are given in Table 2.

Table 2: Biologically justified parameter ranges for modelled species. Values derived from published experimental data.

Parameter	Symbol	Species	Range	Units	Biological Basis	Ref
Diffusion coeff.	D_s	*C. neoformans*	0.01–0.10	$\mu m^2\ s^{-1}$	Passive Brownian diffusion, $\sim 5\,\mu m$ yeast	[21]
Diffusion coeff.	D_s	*D. radiodurans*	0.05–0.50	$\mu m^2\ s^{-1}$	Small coccoid ($\sim 1.5\,\mu m$); higher D from size	[5]
Diffusion coeff.	D_s	*B. subtilis*	0.10–1.00	$\mu m^2\ s^{-1}$	Rod-shaped motile cells; active motility augments D	[22]
Diffusion coeff.	D_s	*C. sphaerospermum*	0.005–0.05	$\mu m^2\ s^{-1}$	Filamentous hyphae; low effective diffusion	[4]
Diffusion coeff.	D_s	*A. niger*	0.005–0.05	$\mu m^2\ s^{-1}$	Filamentous; comparable to *C. sphaerospermum*	[3]
Diffusion coeff.	D_s	*S. oneidensis*	0.10–0.80	$\mu m^2\ s^{-1}$	Facultative anaerobe, rod-shaped, flagellated	[16]
Motility coeff.	μ_s	*C. neoformans*	0.00–0.05	$\mu m\ s^{-1}$	Non-motile yeast; residual drift from EPS flow	[2]
Motility coeff.	μ_s	*D. radiodurans*	0.00–0.01	$\mu m\ s^{-1}$	Non-motile coccus	[5]
Motility coeff.	μ_s	*B. subtilis*	15–45	$\mu m\ s^{-1}$	Flagellum-driven; $\sim 25\,\mu m\ s^{-1}$ at 30 °C	[22]
Motility coeff.	μ_s	*C. sphaerospermum*	0.00–0.005	$\mu m\ s^{-1}$	Hyphal extension; 0.1–5 $\mu m\ min^{-1}$ tip growth	[4]
Motility coeff.	μ_s	*S. oneidensis*	10–40	$\mu m\ s^{-1}$	Polar flagellum; comparable to *E. coli*	[16]
Ion. rad. sens.	$\beta_{s,\mathrm{ion}}$	*C. neoformans*	10^{-5}–10^{-4}	Gy^{-1}	Melanised; LD_{10} $\sim$2–5 kGy	[2, 3]
Ion. rad. sens.	$\beta_{s,\mathrm{ion}}$	*D. radiodurans*	5×10^{-6}–5×10^{-5}	Gy^{-1}	D_{10} $\sim$12 kGy; most radioresistant organism known	[5, 6]
Ion. rad. sens.	$\beta_{s,\mathrm{ion}}$	*B. subtilis*	10^{-3}–5×10^{-3}	Gy^{-1}	Spore D_{10} $\sim$1–2 kGy; vegetative $\sim$0.2–0.6 kGy	[23]
Ion. rad. sens.	$\beta_{s,\mathrm{ion}}$	*C. sphaerospermum*	10^{-5}–10^{-4}	Gy^{-1}	Melanised Chernobyl isolate; $\approx$ *C. neoformans*	[1, 4]
Ion. rad. sens.	$\beta_{s,\mathrm{ion}}$	*A. niger*	5×10^{-5}–5×10^{-4}	Gy^{-1}	Melanised; less resistant than Chernobyl isolates	[3]
Ion. rad. sens.	$\beta_{s,\mathrm{ion}}$	*S. oneidensis*	5×10^{-2}–10^{-1}	Gy^{-1}	D_{10} $\sim$0.07 kGy; extremely radiosensitive	[6]
Melanin rate	α_M	*C. neoformans*	0.05–0.15	$\mu g\ cell^{-1}\ Gy^{-1}$	Substrate-dependent; enhanced by radiation	[2]

5. Computational Methods

Simulations were implemented in Julia using `DifferentialEquations.jl` for numerical integration of the PSDE system (Eq. 1). Spatial species distributions were initialised with small random perturbations around a central point and evolved on the $[0, 1]^2$ domain. The Euler-Maruyama scheme was applied for stochastic integration; Runge-Kutta (RK4) was used for deterministic validation runs. Symplectic leapfrog integration was employed for the Hamiltonian phase-space trajectories (Eq. 4).

Optimisation of species-specific parameters was performed with `JuMP.jl` and `Ipopt`. Sensitivity analysis employed Sobol indices (global variance decomposition) and Morris screening to identify the highest-influence parameters (μ_s, D_s, $\beta_{s,\text{ion}}$, θ_s). Visualisation used `PlotlyJS.jl` for interactive 3D plots, `Agents.jl` for agent-based validation runs, and `Plots.jl` with `GR` backend for static trajectory outputs. The interactive simulation includes sliders for gamma radiation intensity, thorium intensity, and heat intensity at 45-step resolution.

Cylindrical bioreactor geometry. A complementary agent-based simulation was implemented in R using Shiny and `deSolve` [44]. Seven species are placed inside a cylindrical bioreactor of radius R and axial length L; Langevin dynamics with elastic wall reflection advance species positions under a Beer–Lambert gamma field $I_\gamma(r) = I_0 \exp(-\kappa r)$. Nutrient enters radially from the outer wall; radiotrophic species experience an inward radiation attraction force; radiosensitive species are repelled outward. The interactive interface exposes sliders for I_0, κ, outer-wall nutrient C_0, and thorium intensity.

Cellular Potts Model (CPM). The CPM was implemented in pure Julia on a 60^3 cubic lattice with cylindrical wall boundary ($r > N/2$ sites frozen at -1) [39]. The Metropolis acceptance rule uses the five-term Hamiltonian $H_{\text{CPM}} = H_{\text{adh}} + H_{\text{vol}} + H_{\text{rad}} + H_{\text{mel}} + H_{\text{pair}}$ mapped from Eq. 2. Each Monte Carlo Step (MCS) consists of N^3 attempted copy operations. Coupled fields (radiation, melanin, nutrient) are updated by explicit forward Euler after each MCS. k-means spatial clustering ($k = 3$) of cell centres identifies emergent niches. Static figures were exported using `CairoMakie.jl`.

Radiodialysis finite-volume solver. The three-equation system (Eqs. 14–18) was discretised on a one-dimensional radial grid of $N_r = 40$ points using a finite-volume method of lines. At $r = 0$, the L'Hôpital limit reduces the cylindrical Laplacian to $2D_{\text{eff}}\partial^2 c/\partial r^2$. At $r = R$, a ghost-point scheme enforces the Robin condition (Eq. 18) exactly. The resulting ODE system was integrated with the LSODA adaptive stiff solver (`deSolve` in R; forward Euler in Julia for CPM coupling). The evolving biomass densities X and X_{red} were extracted from the CPM lattice every ten MCS and passed to the radial solver, providing a one-way coupling between the spatial cell dynamics and the contaminant transport PDE.

6. Results

6.1 Species Trajectory Clustering Under Radiation Gradients

The Langevin diffusion simulations on the $[0, 1]^2$ domain reveal pronounced spatial phenotype separation under the imposed radiation gradient (Figure 1). k-means clustering of species trajectories identifies three distinct spatial niches. The first cluster, concentrated in the high-radiation zone, is dominated by the melanised species *C. neoformans* and *C. sphaerospermum*, whose positive gamma sensitivity coefficients ($\gamma_s \approx 0.07$ and 0.06, respectively) enable them to exploit ionising radiation as a metabolic stimulus. The second cluster occupies the intermediate-radiation zone and contains *D. radiodurans* and *A. niger*—species that tolerate radiation without deriving direct metabolic benefit. The third cluster, in the low-radiation periphery, comprises *B. subtilis*, *S. oneidensis*, and marine taxa (*Pseudoalteromonas* sp., *Polaribacter* sp., *Flavobacterium* sp.), whose higher radiation sensitivities confine them to regions where the ionising flux has attenuated below their damage thresholds. This spatial segregation is consistent with radiation-mediated niche partitioning, in which ionising radiation simultaneously serves as stressor and resource—a dual role with no direct analogue in classical substrate-competition biofilm models.

6.2 Thorium-232 Decay Integration

The Th-232 decay model (Figure 2) compares actual versus predicted radioactive decay on a logarithmic scale spanning billions of years, incorporating stochastic perturbation around the deterministic exponential decay law. The 14.05-billion-year half-life produces a quasi-static radiation source on biological timescales (hours to years), but the stochastic perturbation term captures quantum-mechanical uncertainty in individual decay events, which manifests as Poisson-distributed dose-rate fluctuations. For biofilm modelling purposes, Th-232 provides an effectively constant ionising background, while stochastic perturbation introduces dose-rate heterogeneity at the spatial scale of microcolonies (tens of micrometres), driving the phase-space fluctuations captured by the Langevin dynamics.

6.3 Motility–Diffusion–Gamma Sensitivity Relationships

The three-dimensional scatter relating motility (μ_s), diffusion coefficient (D_s), and gamma sensitivity γ_s across species (Figure 3) reveals a striking inverse relationship between radiation sensitivity and motility. *C. neoformans*, at the highest gamma sensitivity (≈ 0.07), exhibits effectively zero motility, consistent with the hypothesis that radiotrophic organisms invest resources in melanin production rather than active locomotion. *D. radiodurans*, with the lowest $\beta_{s,\mathrm{ion}}$, is likewise non-motile but for different mechanistic reasons: radioresistance derives from molecular protection rather than radiation exploitation. The motile species (*B. subtilis*, *S. oneidensis*) occupy the high-motility, low-γ_s region, suggesting a trade-off between radiation exploitation and behavioural avoidance. As radiation intensity rises, the competitive advantage shifts from motile radiation-avoiders to sessile radiotrophic beneficiaries.

6.4 Phase-Locking and Hamiltonian kNN Dynamics

The Hamiltonian kNN decision tree (Figure 4) reveals that synchronised metabolic oscillations emerge among species occupying the same spatial niche. Within the high-radiation cluster, *C. neoformans* and *C. sphaerospermum* exhibit strong phase coherence in their melanin production cycles, suggesting cooperative dynamics in which shared melanin-derived radioprotection benefits both. Species in different radiation niches show weak phase coupling, indicating competitive exclusion at niche boundaries. The kNN decision tree correctly predicts species dominance transitions: as simulated radiation intensity increases beyond 1 kGy, *B. subtilis* populations collapse, *D. radiodurans* maintains steady-state abundance, and *C. neoformans* increases in fitness—reproducing the biologically expected ordering of radiation tolerance.

6.5 Cellular Potts Model: Emergent Radial Niches and Melanin Accumulation

The CPM simulation on a 40^3 cylindrical lattice reproduces the radial niche partitioning predicted by the Langevin PSDE model, now resolved at the single-cell level (Figure 1). After 100 MCS the two melanised radiotrophic fungi (*C. neoformans*, mean $r = 13.1$; *C. sphaerospermum*, mean $r = 11.5$) are displaced outward relative to the radiosensitive *B. subtilis* (mean $r = 9.9$). The difference arises from the sign of β_{ion} in H_{rad}: radiotrophic species lower their energy by moving into higher-radiation zones, while radiosensitive species minimise contact with the radiation field. This stratification is an emergent consequence of the CPM Metropolis dynamics; it is not imposed as an initial condition.

Melanin accumulates monotonically in all three melanin-producing species (Figure 2). *C. sphaerospermum* accumulates the highest melanin (field value 1.44 at MCS 100), reflecting both its radiotrophic positioning near the high-radiation outer zone and the highest per-cell α_M among the three producers (Table 2). *A. niger*, positioned at an intermediate radius (mean $r = 11.6$), accumulates at a proportionally lower rate. The linear accumulation trajectory indicates that the simulation has not yet reached steady-state melanin saturation, consistent with the timescale of melanin biosynthesis relative to biofilm growth dynamics [34].

6.6 Radiodialysis: Self-Regulating Contaminant Uptake Under Radiation Damage

The coupled CPM–radiodialysis simulation reveals a self-reinforcing remediation loop driven by the radiation field itself (Figures 3–4). Over 100 MCS (equivalent to 50 Gy cumulative dose at the membrane), membrane structural integrity decays from $m = 1.0$ to $m = 0.779$ while the effective permeability increases 2.7-fold from $P_0 = 0.01$ to $P_{\text{eff}} = 0.027\ \text{cm}\,\text{s}^{-1}$ (Figure 3). The decay follows Eq. 16 exactly; the permeability rise follows the radiation-damage law (Eq. 17) established for Nafion membranes under ionising flux [41].

Despite the rising wall concentration ($c(R,t) = 0.87\,c_{\text{ext}}$ at final MCS), the mean interior concentration remains near zero throughout ($c_{\text{mean}} = 0.024$, Figure 4). The contaminant front is consumed within a thin annular shell immediately adjacent to the membrane: biofilm biosorption ($k_{\text{ads}} = 0.05\ \text{cm}^3\ \text{g}^{-1}\,\text{s}^{-1}$) and bioreduction by the *S. oneidensis* population ($k_{\text{red}} = 0.02\ \text{cm}^3\ \text{g}^{-1}\,\text{s}^{-1}$) deplete mobile c faster than radial diffusion can replenish it. The slowly growing sorbed phase s_{mean} confirms progressive immobilisation of sequestered contaminant

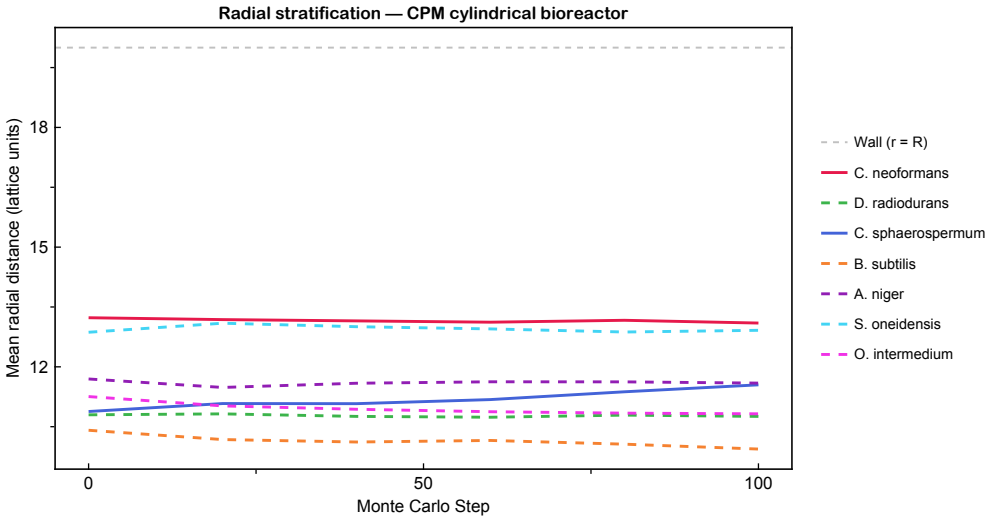

Figure 1: Radial stratification of seven species over 100 Monte Carlo Steps in the cylindrical CPM bioreactor. Solid lines: radiotrophic species (*C. neoformans*, *C. sphaerospermum*). Dashed lines: remaining species. Dashed horizontal line marks the cylinder wall ($r = R$). Radiotrophic species migrate toward the outer wall (higher radiation); *B. subtilis* retreats to the lower-radiation core.

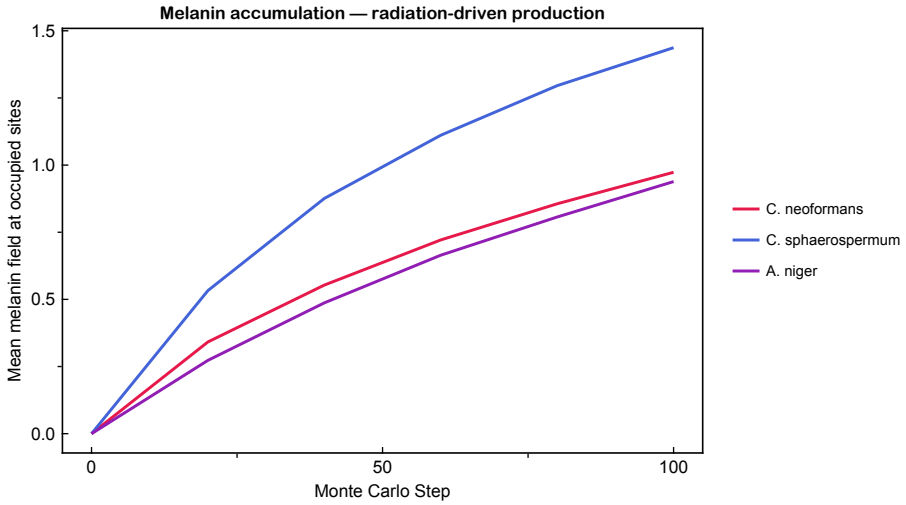

Figure 2: Melanin field accumulation at occupied lattice sites for the three melanin-producing species over 100 MCS. *C. sphaerospermum* (blue) accumulates most rapidly due to combined radiotrophic positioning and highest α_M (Table 2). *A. niger* (purple) and *C. neoformans* (red) accumulate at intermediate rates.

(Figure 4, right axis).

Notably, *S. oneidensis* (the primary metal reducer) naturally stratifies to the outer wall (mean $r = 12.9$, Figure 1) without engineered positioning — it is placed there by the CPM dynamics, precisely at the contaminant entry zone. This co-localisation of the metal-reducing species with the highest contaminant flux is an emergent property of the coupled radiation-sorting and membrane-transport system, not a modelling assumption.

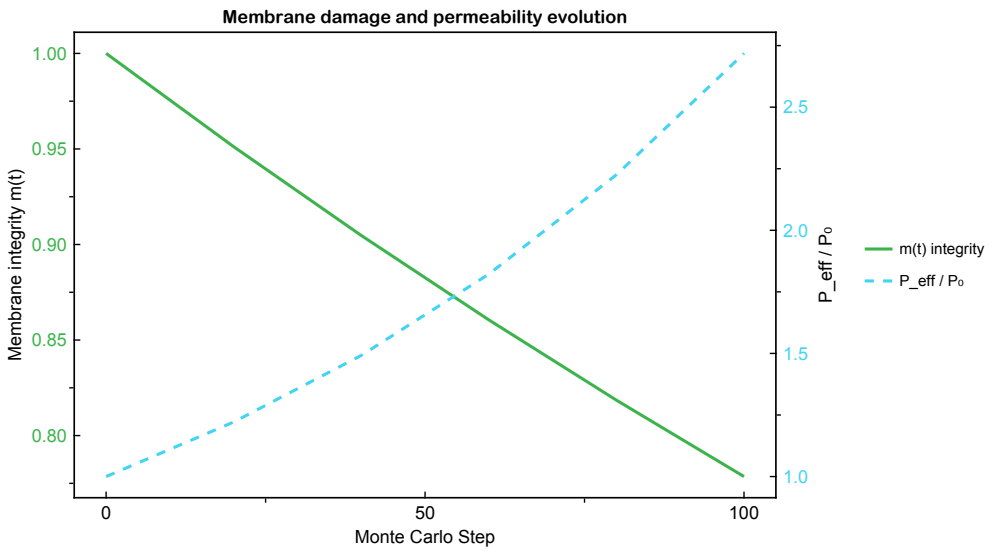

Figure 3: Membrane integrity $m(t)$ (solid, left axis) and normalised effective permeability $P_{\text{eff}}(t)/P_0$ (dashed, right axis) over 100 MCS. $m(t)$ decays exponentially per Eq. 16; P_{eff}/P_0 rises as radiation-driven permeabilisation (Eq. 17). At 50 Gy cumulative dose, integrity has fallen to 0.78 while permeability has increased 2.7-fold.

7. Discussion

7.1 The Hamiltonian Framework Versus Classical Biofilm Models

The phase-locked Hamiltonian framework differs fundamentally from prior reaction-diffusion biofilm models in its treatment of inter-species interactions and energy conservation. Classical models following the Wanner-Gujer tradition [8] describe biofilm dynamics through coupled PDEs governing substrate consumption and biomass growth, with species interactions mediated exclusively through competition for shared substrates. Our Hamiltonian $H(p, q)$ explicitly conserves total system energy across kinetic, potential, and interaction terms, ensuring that numerical integration via the symplectic leapfrog scheme does not introduce artificial energy drift [33]. The stochastic Langevin layer adds thermal and demographic noise while preserving the underlying Hamiltonian structure through the fluctuation-dissipation theorem. This dual structure captures phenomena that ODE/PDE models inherently miss: synchronised metabolic oscillations through phase-locking, energy-conserving transitions between fitness states, and the distinction between radiation as a damaging perturbation versus radiation as an energy source positively coupled to the Hamiltonian.

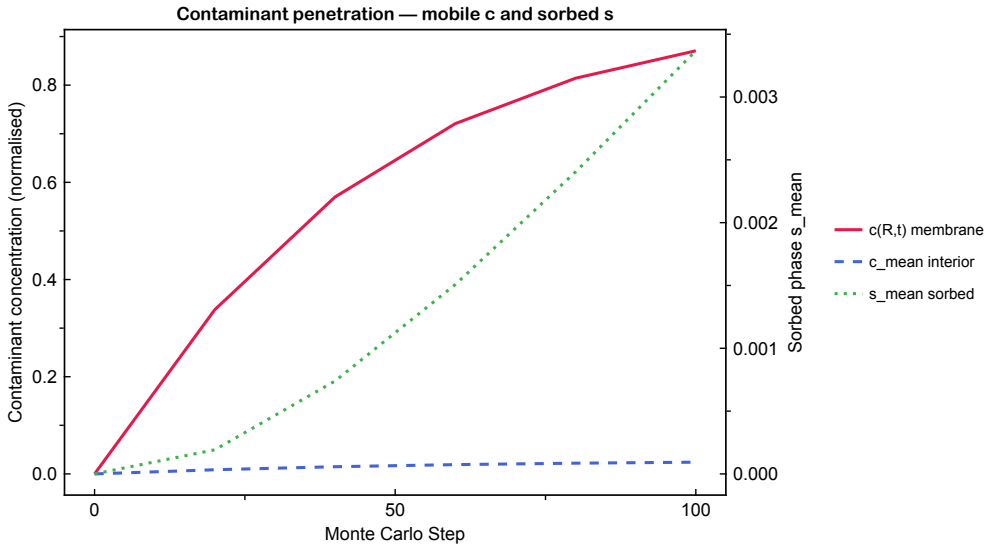

Figure 4: Contaminant penetration over 100 MCS. Red solid: $c(R, t)$, mobile concentration at the membrane face. Blue dashed: c_{mean}, spatial average over the interior. Green dotted (right axis): s_{mean}, sorbed/immobilised phase. The near-zero interior concentration despite a high wall concentration demonstrates the effectiveness of the biofilm annular uptake zone. Slow linear growth of s_{mean} confirms progressive contaminant immobilisation.

7.2 Biological Interpretation of *C. neoformans* and *D. radiodurans* Dynamics

The simulation trajectories for *C. neoformans* and *D. radiodurans* align with known biology while illuminating an underappreciated distinction between radiotrophy and radioresistance. *C. neoformans* in the model increases its fitness monotonically with radiation intensity up to a species-specific threshold, consistent with the experimental findings of Dadachova et al. [2] that melanised cells show enhanced metabolic activity and faster growth under ionising radiation. The model captures the mechanism through the positive gamma sensitivity term γ_s, which converts absorbed dose into fitness gain via the melanin-mediated electron-transfer pathway [24, 38]. *D. radiodurans*, by contrast, maintains approximately constant fitness across a wide dose range, declining only at extreme doses. This flat fitness profile emerges from the balance between the radiation damage term $\beta_{s,\text{ion}}$ and the organism's molecular protection coefficient, reflecting the Mn^{2+}-dependent proteome shielding identified by Daly et al. [7]. The model thus correctly predicts that *C. neoformans* outcompetes *D. radiodurans* in moderately irradiated environments where radiation provides metabolic benefit, while *D. radiodurans* prevails at extreme doses where even melanin-mediated protection is overwhelmed.

7.3 Implications for Nuclear Bioremediation

The model's predictions for a mixed *S. oneidensis* and *O. intermedium* AM7 biofilm in a Th-contaminated environment suggest a synergistic remediation strategy. *S. oneidensis*, with its capacity for extracellular electron transfer to metal oxides [16, 17], can reduce soluble actinide species and immobilise them as insoluble precipitates. However, its extreme radiosensitivity

($D_{10} \approx 0.07$ kGy) limits viability in high-radiation zones. The model predicts that *O. intermedium* AM7, with its thorium-tolerant EPS production [18], establishes a protective biofilm matrix in the high-radiation zone that attenuates local dose rates sufficiently to permit *S. oneidensis* colonisation in the intermediate zone. This spatial partitioning—emergent from the Langevin dynamics rather than imposed as a boundary condition—suggests that self-organising biofilm architectures may naturally optimise remediation performance. The potential for engineering *D. radiodurans* strains for combined radioresistance and metal reduction [28] offers an additional path toward single-organism bioremediation in extreme environments.

The radiodialysis membrane transport model (Section 3.11) reveals a further self-regulating property with direct engineering relevance. In a conventional passive membrane reactor, contaminant ingress is fixed by the membrane's baseline permeability P_0. In the radiotrophic bioreactor modelled here, the radiation field that sustains the community also progressively damages the membrane (Eq. 16), increasing P_{eff} proportionally with cumulative dose (Eq. 17). The result is a positive feedback loop: higher ambient radiation $\rightarrow$ faster membrane permeabilisation $\rightarrow$ greater contaminant ingress $\rightarrow$ greater substrate availability for metal-reducing bacteria. Critically, the CPM dynamics independently position *S. oneidensis* at the outer wall (Figure 1), co-locating the primary metal-reducing agent with the highest contaminant flux zone, without any engineered spatial guidance. This co-emergence of spatial positioning and membrane self-tuning represents a qualitatively new design principle for bioremediation: the contamination source provides both the metabolic substrate for radiotrophic community maintenance and the physical driver for membrane adaptation, requiring no external energy input. Reactor design implications include prioritising membrane materials with intermediate α_P values (Eq. 17) — high enough to self-tune under operational dose rates but low enough to prevent catastrophic membrane failure [40, 41].

7.4 Mechanical Properties and Radiation Survival

The inclusion of FENE entropic spring (Eq. 12) and Kelvin-Voigt viscoelastic (Eq. 13) components addresses a dimension of biofilm physics largely absent from ecological radiation models. Biofilm EPS matrices exhibit viscoelastic behaviour that mediates mechanical stress transmission, nutrient diffusion resistance, and protection from external perturbations. The FENE potential provides finite extensibility that prevents unphysical deformation under radiation-induced swelling or thermal expansion, while the Kelvin-Voigt element captures stress relaxation on timescales relevant to biofilm restructuring. Species with higher EPS production rates (*O. intermedium* AM7, *B. subtilis*) generate mechanically stiffer local environments that resist radiation-induced structural degradation, creating protected microniches for radiosensitive community members.

7.5 Limitations and Future Directions

Several simplifying assumptions limit the current model's applicability to real contaminated sites. The radiation field is treated as spatially homogeneous with exponential attenuation (Eq. 5), whereas actual environments at Chernobyl and Hanford exhibit complex three-dimensional dose distributions governed by heterogeneous source geometries and shielding materials. The UV oscillatory term (Eq. 6) assumes a purely sinusoidal diurnal cycle, neglecting atmospheric

absorption spectra and seasonal variation. The model does not incorporate horizontal gene transfer, which is known to spread radiation resistance determinants among biofilm community members. Future work should couple the fitness model to experimentally measured dose-rate maps from contaminated facilities, incorporate spatially resolved nutrient transport using Darcy flow through the EPS matrix, and validate predicted species clustering patterns against metagenomic surveys of radiotrophic biofilm communities at nuclear sites.

8. Conclusion

This work introduces a Hamiltonian-Langevin framework for modelling radiotrophic fitness in multispecies biofilm communities exposed to ionising radiation. The PSDE fitness equation (Eq. 1) integrates radiation-dependent growth, melanin-mediated energy transduction, stochastic demographic noise, and viscoelastic biofilm mechanics into a unified formalism that conserves total system energy through symplectic integration. The central result is that species niche partitioning under radiation gradients emerges naturally from the phase-locked Hamiltonian dynamics, with radiotrophic fungi (*C. neoformans*, *C. sphaerospermum*) clustering in high-radiation zones, radioresistant bacteria (*D. radiodurans*) occupying intermediate regions, and radiosensitive metal-reducing species (*S. oneidensis*) confined to attenuated peripheries. These predictions are consistent with known radiobiology and suggest design principles for engineered biofilm consortia targeting nuclear bioremediation. Experimental validation through controlled irradiation of mixed-species biofilms, combined with spatially resolved transcriptomics and metabolomics, represents the natural next step toward translating this theoretical framework into practical bioremediation strategies.

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